

PROJECT DOCUMENTATION

LearnMate: Agentic AI for Personalized Course Pathways

Problem Statement No:	Problem Statement No. 12	Academic Track:	Computer Science & Engineering
Project Domain:	EdTech / Artificial Intelligence in Education	Submission Date:	June 8, 2026
Core Technologies:	LangFlow, IBM watsonx.ai, IBM Granite Models, RAG Architecture	Repository Name:	learnmate

1. Executive Summary

In the contemporary digital era, the availability of online educational material has grown exponentially, creating an overwhelming ecosystem for students. Traditional platforms offer generic, linear, and static learning paths that fail to account for a student's existing skill gaps, preferred pace, or long-term career aspirations. This document details the technical layout and functional architecture of **LearnMate**, an intelligent, multi-agent virtual learning coach designed to resolve these deficiencies. Powered by IBM watsonx.ai and orchestrated through LangFlow, LearnMate autonomously evaluates an individual's technical competence, matches their personal interests, and constructs a completely adaptive, dynamic learning pathway using state-of-the-art IBM Granite Large Language Models.

2. Problem Definition & Core Challenges

According to the AICTE programmatic guidelines for Agentic AI implementation, the core problem addressed by Problem Statement 12 encompasses three prominent vectors:

- **Information Overload & Choice Paralysis:** Students face thousands of unstructured course materials, documentation links, and video playlists across various platforms, leading to confusion regarding where to begin or how to progress.
- **Lack of Real-Time Personalization:** Standard educational curricula do not adapt dynamically. If a student already possesses foundational knowledge in a technology (e.g., Vanilla JavaScript), static systems still force them through redundant basic material. Conversely, if they struggle with an advanced concept (e.g., asynchronous event-driven architectures), static courses fail to provide corrective foundational sub-paths.
- **Absence of Continuous Progress Calibration:** Traditional systems lack interactive monitoring, meaning they do not re-route or restructure the remaining syllabus based on milestone feedback or direct assessments.

3. Project Objectives

LearnMate is engineered to transform passive information consumption into a highly contextualized, active mentoring experience. The defined objectives are:

1. To architect an autonomous multi-agent educational pipeline that conducts initial and continuous skill assessments through conversational reasoning.
2. To build a robust Retrieval-Augmented Generation (RAG) framework that references updated tech-stack structures, syllabi, and official documentation repositories.
3. To dynamically output modular, personalized learning roadmaps utilizing enterprise-grade, high-performance language frameworks.
4. To establish a progressive feedback loop capable of updating future milestones based on immediate user performance logs.

4. Proposed Solution & System Architecture

The developed solution implements an end-to-end multi-agent framework built strictly on top of the **LangFlow Orchestration Platform** and powered by **IBM watsonx.ai Enterprise Governance**. Rather than deploying a generic single-prompt chatbot, LearnMate partitions operational duties among four specialized, interacting AI agents:

Agent Entity	Core Operational Responsibility	LangFlow Integration Tooling
Skill Assessment Agent	Conducts conversational dialogue to map the user's current knowledge base, target career goals, and active tech preferences (e.g., UI/UX, Cloud, Frontend, or Full-Stack Node.js/MySQL development).	Chat Input Node, Prompt Template, Custom Memory Buffer
Roadmap Generator Agent	Interacts with the assessment vector logs to structure a step-by-step, milestone-driven technical syllabus optimized to skip redundant topics.	IBM watsonx AI Agent Node, Granite-4.0-8B-Instruct Model
Resource Curator Agent	Utilizes RAG mechanisms to attach official technical documentation, reading lists, and coding assignments to each generated module.	File Component, Vector DB Search Node, Text Splitters
Progress Tracking Agent	Evaluates user submissions or direct chat logs to confirm concept mastery, modifying the remaining timeline dynamically if roadblocks are encountered.	Prompt Logic gates, Conditional Flow Routing, Chat Output

Architectural Mechanics: The multi-agent workflow operates in a linear-feedback topology. The data ingested by the Chat Input passes through systemic prompt nodes to constrain the behavioral envelope of the IBM Granite underlying model, ensuring responses remain pedagogically professional, accurate, and structurally organized into JSON-parsable milestone arrays.

5. Technical Stack & Implementation Details

To ensure complete alignment with the evaluation criteria of the IBM SkillsBuild University Engagement track, the following technical components were utilized:

- **Orchestration Platform:** LangFlow (v1.0+) visual canvas layout for deterministic and agentic workflow wiring.
- **AI Model Provider:** IBM watsonx.ai Foundation Models.
- **Core LLM Base:** Granite-4.0-8B-Instruct / ibm-granite-3-2-8b, selected due to their superior performance in technical reasoning, structured layout generation, and strict instruction-following capabilities.
- **RAG Pipeline Components:** File Loader nodes coupled with mathematical text splitters, embedded via IBM embeddings and mapped using localized high-performance Vector indices for semantic syllabus alignment.
- **Interface Layer (Future Extension Code):** Python Streamlit infrastructure mapping real-time communication protocols with the backend LangFlow execution endpoints.

6. Novelty, Key Differentiators & Future Scope

The LearnMate platform completely redefines online learning by introducing an active, self-correcting mentorship element. Traditional platforms serve static video links; LearnMate behaves like a live human supervisor. Its unique value lies in its **hyper-personalized routing capability**, ensuring zero time is wasted on known concepts while expanding focus on areas that require additional support.

Future Roadmap: The modular layout of the platform allows for immediate horizontal scaling. Future variations will focus on integration into institutional Learning Management Systems (LMS) for long-term tracking and cognitive monitoring. Furthermore, it can be extended via Developer IDE APIs (such as custom extensions for VS Code) to directly observe syntax architecture and logic-building, delivering tailored documentation references the exact second a student encounters a compiler error or architectural blocker.